

EXERCISE 3A

Solution 1:

$$\begin{aligned} &9x^2 + 12xy \\ &= 3x(3x + 4y) \end{aligned}$$

Solution 2:

$$\begin{aligned} &18x^2y - 24xyz \\ &= 6xy(3x - 4z) \end{aligned}$$

Solution 3:

$$\begin{aligned} &27a^3b^3 - 45a^4b^2 \\ &= 9a^3b^2(3b - 5a) \end{aligned}$$

Solution 4:

$$\begin{aligned} &2a(x + y) - 3b(x + y) \\ &= (x + y)(2a - 3b) \end{aligned}$$

Solution 5:

$$\begin{aligned} &2x(p^2 + q^2) + 4y(p^2 + q^2) \\ &= (p^2 + q^2)(2x + 4y) \\ &= 2(p^2 + q^2)(x + 2y) \end{aligned}$$

Solution 6:

$$\begin{aligned} &x(a - 5) + y(5 - a) \\ &= x(a - 5) - y(a - 5) \\ &= (a - 5)(x - y) \end{aligned}$$

Solution 7:

$$\begin{aligned} &4(a + b) - 6(a + b)^2 \\ &= (a + b)\{4 - 6(a + b)\} \\ &= (a + b)(4 - 6a - 6b) \\ &= 2(a + b)(2 - 3a - 3b) \end{aligned}$$

Solution 8:

$$\begin{aligned} & 8(3a - 2b)^2 - 10(3a - 2b) \\ &= 2(3a - 2b)\{4(3a - 2b) - 5\} \\ &= 2(3a - 2b)(12a - 8b - 5) \end{aligned}$$

Solution 9:

$$\begin{aligned} & x(x + y)^3 - 3x^2y(x + y) \\ &= x(x + y)\{(x + y)^2 - 3xy\} \\ &= x(x + y)\{x^2 + y^2 + 2xy - 3xy\} \\ &= x(x + y)(x^2 + y^2 - xy) \end{aligned}$$

Solution 10:

$$\begin{aligned} & x^3 + 2x^2 + 5x + 10 \\ &= x^2(x + 2) + 5(x + 2) \\ &= (x + 2)(x^2 + 5) \end{aligned}$$

Solution 11:

$$\begin{aligned} & x^2 + xy - 2xz - 2yz \\ &= x(x + y) - 2z(x + y) \\ &= (x + y)(x - 2z) \end{aligned}$$

Solution 12:

$$\begin{aligned} & a^3b - a^2b + 5ab - 5b \\ &= a^2b(a - 1) + 5b(a - 1) \\ &= (a - 1)(a^2b + 5b) \\ &= b(a - 1)(a^2 + 5) \end{aligned}$$

Solution 13:

$$\begin{aligned} & 8 - 4a - 2a^3 + a^4 \\ &= 4(2 - a) - a^3(2 - a) \\ &= (2 - a)(4 - a^3) \end{aligned}$$

Solution 14:

$$\begin{aligned}x^3 - 2x^2y + 3xy^2 - 6y^3 \\&= x^2(x - 2y) + 3y^2(x - 2y) \\&= (x - 2y)(x^2 + 3y^2)\end{aligned}$$

Solution 15:

$$\begin{aligned}px - 5q + pq - 5x \\&= px + pq - 5x - 5q \\&= p(x + q) - 5(x + q) \\&= (x + q)(p - 5)\end{aligned}$$

Solution 16:

$$\begin{aligned}x^2 + y - xy - x \\&= x^2 - xy - x + y \\&= x(x - y) - (x - y) \\&= (x - y)(x - 1)\end{aligned}$$

Solution 17:

$$\begin{aligned}(3a - 1)^2 - 6a + 2 \\&= (3a - 1)^2 - 2(3a - 1) \\&= (3a - 1)\{(3a - 1) - 2\} \\&= (3a - 1)(3a - 3) \\&= 3(3a - 1)(a - 1)\end{aligned}$$

Solution 18:

$$\begin{aligned}(2x - 3)^2 - 8x + 12 \\&= (2x - 3)^2 - 4(2x - 3) \\&= (2x - 3)\{(2x - 3) - 4\} \\&= (2x - 3)(2x - 7)\end{aligned}$$

Solution 19:

$$\begin{aligned}a^3 + a - 3a^2 - 3 \\&= a^3 - 3a^2 + a - 3 \\&= a^2(a - 3) + (a - 3) \\&= (a - 3)(a^2 + 1)\end{aligned}$$

Solution 20:

$$\begin{aligned} & 3ax - 6ay - 8by + 4bx \\ &= 3ax - 6ay + 4bx - 8by \\ &= 3a(x - 2y) + 4b(x - 2y) \\ &= (x - 2y)(3a + 4b) \end{aligned}$$

Solution 21:

$$\begin{aligned} & abx^2 + a^2x + b^2x + ab \\ &= ax(bx + a) + b(bx + a) \\ &= (bx + a)(ax + b) \end{aligned}$$

Solution 22:

$$\begin{aligned} & x^3 - x^2 + ax + x - a - 1 \\ &= x^2(x - 1) + ax - a + x - 1 \\ &= x^2(x - 1) + a(x - 1) + (x - 1) \\ &= (x - 1)(x^2 + a + 1) \end{aligned}$$

Solution 23:

$$\begin{aligned} & 2x + 4y - 8xy - 1 \\ &= 2x - 8xy + 4y - 1 \\ &= 2x(1 - 4y) - (1 - 4y) \\ &= (1 - 4y)(2x - 1) \end{aligned}$$

Solution 24:

$$\begin{aligned} & ab(x^2 + y^2) - xy(a^2 + b^2) \\ &= abx^2 + aby^2 - xya^2 - xyb^2 \\ &= abx^2 - a^2xy - b^2xy + aby^2 \\ &= ax(bx - ay) - by(bx - ay) \\ &= (bx - ay)(ax - by) \end{aligned}$$

Solution 25:

$$\begin{aligned} & a^2 + ab^2 + ab + b^3 \\ &= a^2 + ab + ab^2 + b^3 \\ &= a(a + b) + b^2(a + b) \end{aligned}$$

$$= (a + b)(a + b^2)$$

Solution 26:

$$\begin{aligned} & a^3 + ab(1 - 2a) - 2b^2 \\ &= a^3 + ab - 2a^2b - 2b^2 \\ &= a^3 - 2a^2b + ab - 2b^2 \\ &= a^2(a - 2b) + b(a - 2b) \\ &= (a - 2b)(a^2 + b) \end{aligned}$$

Solution 27:

$$\begin{aligned} & 2a^2 + bc - 2ab - ac \\ &= (2a^2 - 2ab) - (ac - bc) \\ &= 2a(a - b) - c(a - b) \\ &= (a - b)(2a - c) \end{aligned}$$

Solution 28:

$$\begin{aligned} & (ax + by)^2 + (bx - ay)^2 \\ &= (ax)^2 + (by)^2 + 2abxy + (bx)^2 + (ay)^2 - 2abxy \\ &= a^2x^2 + b^2y^2 + b^2x^2 + a^2y^2 \\ &= a^2x^2 + b^2x^2 + b^2y^2 + a^2y^2 \\ &= x^2(a^2 + b^2) + y^2(a^2 + b^2) \\ &= (a^2 + b^2)(x^2 + y^2) \end{aligned}$$

Solution 29:

$$\begin{aligned} & a(a + b - c) - bc \\ &= a^2 + ab - ac - bc \\ &= a^2 - ac + ab - bc \\ &= a(a - c) + b(a - c) \\ &= (a - c)(a + b) \end{aligned}$$

Solution 30:

$$\begin{aligned} & a(a - 2b - c) + 2bc \\ &= a^2 - 2ab - ac + 2bc \\ &= a(a - 2b) - c(a - 2b) \end{aligned}$$

$$= (a - 2b)(a - c)$$

Solution 31:

$$a^2x^2 + (ax^2 + 1)x + a$$

$$= a^2x^2 + ax^3 + x + a$$

$$= ax^2(a + x) + (x + a)$$

$$= (a + x)(ax^2 + 1)$$

Solution 32:

$$ab(x^2 + 1) + x(a^2 + b^2)$$

$$= abx^2 + ab + a^2x + b^2x$$

$$= abx^2 + a^2x + ab + b^2x$$

$$= ax(bx + a) + b(bx + a)$$

$$= (ax + b)(bx + a)$$

Solution 33:

$$x^2 - (a + b)x + ab$$

$$= x^2 - ax - bx + ab$$

$$= x(x - a) - b(x - a)$$

$$= (x - a)(x - b)$$

Solution 34:

$$x^2 + \frac{1}{x^2} - 2 - 3x + \frac{3}{x}$$

$$= x^2 - 2 + \frac{1}{x^2} - 3x + \frac{3}{x}$$

$$= x^2 - 2 \times x \times \frac{1}{x} + \left(\frac{1}{x}\right)^2 - 3\left(x - \frac{1}{x}\right)$$

$$= \left(x - \frac{1}{x}\right)^2 - 3\left(x - \frac{1}{x}\right)$$

$$= \left(x - \frac{1}{x}\right)\left\{x - \frac{1}{x} - 3\right\}$$

EXERCISE 3B**Solution 1:**

$$\begin{aligned} & 9x^2 - 16y^2 \\ &= (3x)^2 - (4y)^2 \\ &= (3x + 4y)(3x - 4y) \quad [a^2 - b^2 = (a + b)(a - b)] \end{aligned}$$

Solution 2:

$$\begin{aligned} & \left(\frac{25}{4}x^2 - \frac{1}{9}y^2\right) \\ &= \left(\frac{5}{2}x\right)^2 - \left(\frac{1}{3}y\right)^2 \\ &= \left(\frac{5}{2}x - \frac{1}{3}y\right)\left(\frac{5}{2}x + \frac{1}{3}y\right) \quad [a^2 - b^2 = (a + b)(a - b)] \end{aligned}$$

Solution 3:

$$\begin{aligned} & 81 - 16x^2 \\ &= 9^2 - (4x)^2 \\ &= (9 + 4x)(9 - 4x) \quad [a^2 - b^2 = (a + b)(a - b)] \end{aligned}$$

Solution 4:

$$\begin{aligned} & 5 - 20x^2 \\ &= 5(1 - 4x^2) \\ &= 5\{1^2 - (2x)^2\} \\ &= 5(1 + 2x)(1 - 2x) \quad [a^2 - b^2 = (a + b)(a - b)] \end{aligned}$$

Solution 5:

$$\begin{aligned} & 2x^4 - 32 \\ &= 2(x^4 - 16) \\ &= 2\{(x^2)^2 - 4^2\} \\ &= 2\{(x^2 + 4)(x^2 - 4)\} \\ &= 2(x^2 + 4)\{x^2 - 2^2\} \end{aligned}$$

$$= 2(x^2 + 4)(x + 2)(x - 2)$$

$$[a^2 - b^2 = (a + b)(a - b)]$$

Solution 6:

$$\begin{aligned} & 3a^3b - 243ab^3 \\ &= 3ab(a^2 - 81b^2) \\ &= 3ab\{a^2 - (9b)^2\} \\ &= 3ab(a + 9b)(a - 9b) \end{aligned}$$

Solution 7:

$$\begin{aligned} & 3x^3 - 48x \\ &= 3x(x^2 - 16) \\ &= 3x\{x^2 - 4^2\} \\ &= 3x(x + 4)(x - 4) \end{aligned}$$

Solution 8:

$$\begin{aligned} & 27a^2 - 48b^2 \\ &= 3(9a^2 - 16b^2) \\ &= 3\{(3a)^2 - (4b)^2\} \\ &= 3(3a + 4b)(3a - 4b) \end{aligned}$$

Solution 9:

$$\begin{aligned} & x - 64x^3 \\ &= x(1 - 64x^2) \\ &= x\{1^2 - (8x)^2\} \\ &= x(1 + 8x)(1 - 8x) \end{aligned}$$

Solution 10:

$$\begin{aligned} & 8ab^2 - 18a^3 \\ &= 2a(4b^2 - 9a^2) \\ &= 2a\{(2b)^2 - (3a)^2\} \end{aligned}$$

$$= 2a(2b - 3a)(2b + 3a)$$

Solution 11:

$$\begin{aligned} 150 - 6x^2 \\ &= 6(25 - x^2) \\ &= 6\{5^2 - x^2\} \\ &= 6(5 + x)(5 - x) \end{aligned}$$

Solution 12:

$$\begin{aligned} 2 - 50x^2 \\ &= 2(1 - 25x^2) \\ &= 2\{1^2 - (5x)^2\} \\ &= 2(1 + 5x)(1 - 5x) \end{aligned}$$

Solution 13:

$$\begin{aligned} 20x^2 - 45 \\ &= 5(4x^2 - 9) \\ &= 5\{(2x)^2 - 3^2\} \\ &= 5(2x + 3)(2x - 3) \end{aligned}$$

Solution 14:

$$\begin{aligned} (3a + 5b)^2 - 4c^2 \\ &= (3a + 5b)^2 - (2c)^2 \\ &= \{(3a + 5b + 2c)(3a + 5b - 2c)\} \end{aligned}$$

Solution 15:

$$\begin{aligned} a^2 - b^2 - a - b \\ &= (a + b)(a - b) - (a + b) \\ &= (a + b)\{(a - b) - 1\} \\ &= (a + b)(a - b - 1) \end{aligned}$$

Solution 16:

$$\begin{aligned} & 4a^2 - 9b^2 - 2a - 3b \\ &= (2a)^2 - (3b)^2 - (2a + 3b) \\ &= (2a + 3b)(2a - 3b) - (2a + 3b) \\ &= (2a + 3b)\{(2a - 3b) - 1\} \\ &= (2a + 3b)\{2a - 3b - 1\} \end{aligned}$$

Solution 17:

$$\begin{aligned} & a^2 - b^2 + 2bc - c^2 \\ &= a^2 - (b^2 - 2bc + c^2) \\ &= a^2 - (b - c)^2 \\ &= (a + b - c)\{a - (b - c)\} \\ &= (a + b - c)(a - b + c) \end{aligned}$$

Solution 18:

$$\begin{aligned} & 4a^2 - 4b^2 + 4a + 1 \\ &= 4a^2 + 4a + 1 - 4b^2 \\ &= \{(2a)^2 + 2 \times 2a \times 1 + 1^2\} - (2b)^2 \\ &= (2a + 1)^2 - (2b)^2 \\ &= (2a + 1 - 2b)(2a + 1 + 2b) \\ &= (2a + 2b + 1)(2a - 2b + 1) \end{aligned}$$

Solution 19:

$$\begin{aligned} & a^2 + 2ab + b^2 - 9c^2 \\ &= (a + b)^2 - (3c)^2 \\ &= (a + b + 3c)(a + b - 3c) \end{aligned}$$

Solution 20:

$$108a^2 - 3(b - c)^2$$

$$\begin{aligned} &= 3\{36a^2 - (b - c)^2\} \\ &= 3\{(6a)^2 - (b - c)^2\} \\ &= 3(6a - b + c)(6a + b - c) \end{aligned}$$

Solution 21:

$$\begin{aligned} &(a + b)^3 - a - b \\ &= (a + b)^3 - (a + b) \\ &= (a + b)\{(a + b)^2 - 1\} \\ &= (a + b)\{(a + b)^2 - 1^2\} \\ &= (a + b)(a + b + 1)(a + b - 1) \end{aligned} \quad \because [a^2 - b^2 = (a + b)(a - b)]$$

Solution 22:

$$\begin{aligned} &x^2 + y^2 - z^2 - 2xy \\ &= x^2 - 2xy + y^2 - z^2 \\ &= (x - y)^2 - z^2 \\ &= (x - y + z)(x - y - z) \end{aligned}$$

Solution 23:

$$\begin{aligned} &x^2 + 2xy + y^2 - a^2 + 2ab - b^2 \\ &= (x + y)^2 - (a^2 - 2ab + b^2) \\ &= (x + y)^2 - (a - b)^2 \\ &= \{(x + y) + (a - b)\}\{(x + y) - (a - b)\} \\ &= (x + y + a - b)(x + y - a + b) \end{aligned}$$

Solution 24:

$$\begin{aligned} &25x^2 - 10x + 1 - 36y^2 \\ &= (5x)^2 - 2 \times 5x \times 1 + 1^2 - (6y)^2 \\ &= (5x - 1)^2 - (6y)^2 \\ &= (5x - 1 + 6y)(5x - 1 - 6y) \end{aligned}$$

Solution 25:

$$\begin{aligned} & a - b - a^2 + b^2 \\ &= (a - b) - (a^2 - b^2) \\ &= (a - b) - (a + b)(a - b) \\ &= (a - b)\{1 - (a + b)\} \\ &= (a - b)(1 - a - b) \end{aligned}$$

Solution 26:

$$\begin{aligned} & a^2 - b^2 - 4ac + 4c^2 \\ &= a^2 - 4ac + 4c^2 - b^2 \\ &= a^2 - 2 \times a \times 2c + (2c)^2 - b^2 \\ &= (a - 2c)^2 - b^2 \\ &= (a - 2c + b)(a - 2c - b) \end{aligned}$$

Solution 27:

$$\begin{aligned} & 9 - a^2 + 2ab - b^2 \\ &= 9 - (a^2 - 2ab + b^2) \\ &= 3^2 - (a - b)^2 \\ &= (3 + a - b)(3 - a + b) \end{aligned}$$

Solution 28:

$$\begin{aligned} & x^3 - 5x^2 - x + 5 \\ &= x^2(x - 5) - (x - 5) \\ &= (x - 5)(x^2 - 1) \\ &= (x - 5)(x + 1)(x - 1) \end{aligned}$$

Solution 29:

$$\begin{aligned} & 1 + 2ab - (a^2 + b^2) \\ &= 1 + 2ab - a^2 - b^2 \\ &= 1 - a^2 + 2ab - b^2 \end{aligned}$$

$$\begin{aligned} &= 1 - (a^2 - 2ab + b^2) \\ &= 1^2 - (a - b)^2 \\ &= (1 + a - b)(1 - a + b) \end{aligned}$$

Solution 30:

$$\begin{aligned} &9a^2 + 6a + 1 - 36b^2 \\ &= (3a)^2 + 2 \times 3a \times 1 + 1^2 - (6b)^2 \\ &= (3a + 1)^2 - (6b)^2 \\ &= (3a + 1 + 6b)(3a + 1 - 6b) \\ &= (3a + 6b + 1)(3a - 6b + 1) \end{aligned}$$

Solution 31:

$$\begin{aligned} &x^2 - y^2 + 6y - 9 \\ &= x^2 - (y^2 - 6y + 9) \\ &= x^2 - (y - 3)^2 \\ &= (x + y - 3)\{x - (y - 3)\} \\ &= (x + y - 3)(x - y + 3) \end{aligned}$$

Solution 32:

$$\begin{aligned} &4x^2 - 9y^2 - 2x - 3y \\ &= (2x)^2 - (3y)^2 - (2x + 3y) \\ &= (2x + 3y)(2x - 3y) - (2x + 3y) \\ &= (2x + 3y)\{2x - 3y - 1\} \end{aligned}$$

Solution 33:

$$\begin{aligned} &9a^2 + 3a - 8b - 64b^2 \\ &= (3a)^2 - (8b)^2 + (3a - 8b) \\ &= (3a + 8b)(3a - 8b) + (3a - 8b) \\ &= (3a - 8b)\{3a + 8b + 1\} \end{aligned}$$

Solution 34:

$$\begin{aligned} & x^2 + \frac{1}{x^2} - 3 \\ &= x^2 - 2 + \frac{1}{x^2} - 1 \\ &= \left[x^2 - 2 \times x^2 \times \frac{1}{x^2} + \left(\frac{1}{x} \right)^2 \right] - 1 \\ &= \left(x - \frac{1}{x} \right)^2 - 1^2 \\ &= \left(x - \frac{1}{x} + 1 \right) \left(x - \frac{1}{x} - 1 \right) \end{aligned}$$

Solution 35:

$$\begin{aligned} & x^2 - 2 + \frac{1}{x^2} - y^2 \\ &= \left[x^2 - 2 \times x \times \frac{1}{x} + \frac{1}{x^2} \right] - y^2 \\ &= \left(x - \frac{1}{x} \right)^2 - y^2 \\ &= \left(x - \frac{1}{x} + y \right) \left(x - \frac{1}{x} - y \right) \end{aligned}$$

Solution 36:

$$\begin{aligned} & x^4 + \frac{4}{x^4} \\ &= x^4 + \frac{4}{x^4} + 4 - 4 \\ &= \left[(x^2)^2 + 2 \times x^2 \times \frac{2}{x^2} + \left(\frac{2}{x^2} \right)^2 \right] - 2^2 \\ &= \left(x^2 + \frac{2}{x^2} \right)^2 - 2^2 \\ &= \left(x^2 + \frac{2}{x^2} + 2 \right) \left(x^2 + \frac{2}{x^2} - 2 \right) \end{aligned}$$

Solution 37:

$$\begin{aligned}x^8 - 1 &= (x^4)^2 - 1^2 \\&= [(x)^2 - (1)^2](x^4 + 1) \\&= (x^2 - 1)(x^2 + 1)(x^4 + 1) \\&= (x - 1)(x + 1)(x^2 + 1)(x^4 + 1) \\&= (x - 1)(x + 1)(x^2 + 1)[(x^2)^2 + (1)^2] \\&= (x - 1)(x + 1)(x^2 + 1)[(x^2)^2 + (1)^2 + 2x^2 - 2x^2] \\&= (x - 1)(x + 1)(x^2 + 1)[(x^2 + 1)^2 - 2x^2] \\&= (x - 1)(x + 1)(x^2 + 1)[(x^2 + 1)^2 - (\sqrt{2}x)^2] \\&= (x - 1)(x + 1)(x^2 + 1)(x^2 + 1 - \sqrt{2}x)(x^2 + 1 + \sqrt{2}x)\end{aligned}$$

Solution 38:

$$\begin{aligned}16x^4 - 1 &= (4x^2)^2 - 1^2 \\&= (4x^2 + 1)(4x^2 - 1) \\&= (4x^2 + 1)[(2x)^2 - 1^2] \\&= (4x^2 + 1)(2x + 1)(2x - 1)\end{aligned}$$

Solution 39:

$$\begin{aligned}81x^4 - y^4 &= (9x^2)^2 - (y^2)^2 \\&= (9x^2 - y^2)(9x^2 + y^2) \\&= [(3x)^2 - y^2](9x^2 + y^2) \\&= (3x + y)(3x - y)(9x^2 + y^2)\end{aligned}$$

Solution 40: $x^4 - 625$

$$\begin{aligned}&= (x^2)^2 - 25^2 \\&= (x^2 - 25)(x^2 + 25)\end{aligned}$$

$$= (x^2 - 5^2)(x^2 + 25)$$

$$= (x + 5)(x - 5)(x^2 + 25)$$

EXERCISE 3C

Solution 1:

$$x^2 + 11x + 30$$

We split 11 into two numbers such that their sum is 11 and their product is 30

$$= x^2 + (5 + 6)x + 30$$

$$= x^2 + 5x + 6x + 30$$

$$= (x + 5)(x + 6)$$

Solution 2:

$$x^2 + 18x + 32$$

We split 18 into two numbers such that their sum 18 and their product is 32.

Clearly $16 + 2 = 18$ and $16 \times 2 = 32$

$$= x^2 + 16x + 2x + 32$$

$$= x(x + 16) + 2(x + 16)$$

$$= (x + 16)(x + 2)$$

Solution 3:

$$x^2 + 20x - 69$$

We split 20 into two numbers such that their sum 20 and their product is 69.

Clearly $23 - 3 = 20$ and $23 \times 3 = 69$

$$= x^2 + 23x - 3x - 69$$

$$= x(x + 23) - 3(x + 23)$$

$$= (x + 23)(x - 3)$$

Solution 4:

$$x^2 + 19x - 150$$

$$= x^2 + 25x - 6x - 150$$

$$= x(x + 25) - 6(x + 25)$$

$$= (x + 25)(x - 6)$$

Solution 5:

$$\begin{aligned}x^2 + 7x - 98 \\&= x^2 + 14x - 7x - 98 \\&= x(x + 14) - 7(x + 14) \\&= (x + 14)(x - 7)\end{aligned}$$

Solution 6:

$$\begin{aligned}x^2 + 2\sqrt{3}x - 24 \\&= x^2 + 4\sqrt{3}x - 2\sqrt{3}x - 24 \\&= x(x + 4\sqrt{3}) - 2\sqrt{3}(x + 4\sqrt{3}) \\&= (x + 4\sqrt{3})(x - 2\sqrt{3})\end{aligned}$$

Solution 7:

$$\begin{aligned}x^2 - 21x + 90 \\&= x^2 - 15x - 6x + 90 \\&= x(x - 15) - 6(x - 15) \\&= (x - 15)(x - 6)\end{aligned}$$

Solution 8:

$$\begin{aligned}x^2 - 22x + 120 \\&= x^2 - 12x - 10x + 120 \\&= x(x - 12) - 10(x - 12) \\&= (x - 12)(x - 10)\end{aligned}$$

Solution 9:

$$\begin{aligned}x^2 - 4x + 3 \\&= x^2 - 3x - x + 3 \\&= x(x - 3) - 1(x - 3) \\&= (x - 3)(x - 1)\end{aligned}$$

Solution 10:

$$\begin{aligned} & x^2 + 7\sqrt{6}x + 60 \\ &= x^2 + 5\sqrt{6}x + 2\sqrt{6}x + 60 \\ &= x(x + 5\sqrt{6}) + 2\sqrt{6}(x + 5\sqrt{6}) \\ &= (x + 5\sqrt{6})(x + 2\sqrt{6}) \end{aligned}$$

Solution 11:

$$\begin{aligned} & x^2 + 3\sqrt{3}x + 6 \\ &= x^2 + 2\sqrt{3}x + \sqrt{3}x + 6 \\ &= x(x + 2\sqrt{3}) + \sqrt{3}(x + 2\sqrt{3}) \\ &= (x + 2\sqrt{3})(x + \sqrt{3}) \end{aligned}$$

Solution 12:

$$\begin{aligned} & x^2 + 6\sqrt{6}x + 48 \\ &= x^2 + 4\sqrt{6}x + 2\sqrt{6}x + 48 \\ &= x(x + 4\sqrt{6}) + 2\sqrt{6}(x + 4\sqrt{6}) \\ &= (x + 4\sqrt{6})(x + 2\sqrt{6}) \end{aligned}$$

Solution 13:

$$\begin{aligned} & x^2 + 5\sqrt{5}x + 30 \\ &= x^2 + 3\sqrt{5}x + 2\sqrt{5}x + 30 \\ &= x(x + 3\sqrt{5}) + 2\sqrt{5}(x + 3\sqrt{5}) \quad [\because 30 = 2\sqrt{5} \times 3\sqrt{5}] \\ &= (x + 3\sqrt{5})(x + 2\sqrt{5}) \end{aligned}$$

Solution 14:

$$\begin{aligned} & x^2 - 24x - 180 \\ &= x^2 - 30x + 6x - 180 \\ &= x(x - 30) + 6(x - 30) \end{aligned}$$

$$= (x - 30)(x + 6)$$

Solution 15:

$$x^2 - 32x - 105$$

$$= x^2 - 35x + 3x - 105$$

$$= x(x - 35) + 3(x - 35)$$

$$= (x - 35)(x + 3)$$

Solution 16:

$$x^2 - 11x - 80$$

$$= x^2 - 16x + 5x - 80$$

$$= x(x - 16) + 5(x - 16)$$

$$= (x - 16)(x + 5)$$

Solution 17:

$$6 - x - x^2$$

$$= 6 - 3x + 2x - x^2$$

$$= 3(2 - x) + x(2 - x)$$

$$= (2 - x)(3 + x)$$

Solution 18:

$$x^2 - \sqrt{3}x - 6$$

$$= x^2 - 2\sqrt{3}x + \sqrt{3}x - 6$$

$$= x(x - 2\sqrt{3}) + \sqrt{3}(x - 2\sqrt{3})$$

$$= (x - 2\sqrt{3})(x + \sqrt{3})$$

Solution 19:

$$40 + 3x - x^2$$

$$= 40 + 8x - 5x - x^2$$

$$= 8(5 + x) - x(5 + x)$$

$$= (5 + x)(8 - x)$$

Solution 20:

$$x^2 - 26x + 133$$

$$= x^2 - 19x - 7x + 133$$

$$= x(x - 19) - 7(x - 19)$$

$$= (x - 19)(x - 7)$$

Solution 21:

$$x^2 - 2\sqrt{3}x - 24$$

$$= x^2 - 4\sqrt{3}x + 2\sqrt{3}x - 24$$

$$= x(x - 4\sqrt{3}) + 2\sqrt{3}(x - 4\sqrt{3})$$

$$= (x - 4\sqrt{3})(x + 2\sqrt{3})$$

Solution 22:

$$x^2 - 3\sqrt{5}x - 20$$

$$= x^2 - 4\sqrt{5}x + \sqrt{5}x - 20$$

$$= x(x - 4\sqrt{5}) + \sqrt{5}(x - 4\sqrt{5})$$

$$= (x - 4\sqrt{5})(x + \sqrt{5})$$

Solution 23:

$$x^2 + \sqrt{2}x - 24$$

$$= x^2 + 4\sqrt{2}x - 3\sqrt{2}x - 24$$

$$= x(x + 4\sqrt{2}) - 3\sqrt{2}(x + 4\sqrt{2})$$

$$= (x + 4\sqrt{2})(x - 3\sqrt{2})$$

Solution 24:

$$x^2 - 2\sqrt{2}x - 30$$

$$\begin{aligned} &= x^2 - 5\sqrt{2}x + 3\sqrt{2}x - 30 \\ &= x(x - 5\sqrt{2}) + 3\sqrt{2}(x - 5\sqrt{2}) \\ &= (x - 5\sqrt{2})(x + 3\sqrt{2}) \end{aligned}$$

Solution 25:

$$\begin{aligned} &x^2 - x - 156 \\ &= x^2 - 13x + 12x - 156 \\ &= x(x - 13) + 12(x - 13) \\ &= (x - 13)(x + 12) \end{aligned}$$

Solution 26:

$$\begin{aligned} &x^2 - 24x - 145 \\ &= x^2 + 5x - 29x - 145 \\ &= x(x + 5) - 29(x + 5) \\ &= (x - 29)(x + 5) \end{aligned}$$

Solution 27:

$$\begin{aligned} &9x^2 + 18x + 8 \\ &= 9x^2 + 12x + 6x + 8 \\ &= 3x(3x + 4) + 2(3x + 4) \\ &= (3x + 4)(3x + 2) \end{aligned}$$

Solution 28:

$$\begin{aligned} &6x^2 + 17x + 12 \\ &= 6x^2 + 9x + 8x + 12 \\ &= 3x(2x + 3) + 4(2x + 3) \\ &= (2x + 3)(3x + 4) \end{aligned}$$

Solution 29:

$$18x^2 + 3x - 10$$

We have to split 3 into two numbers such that their sum is 3 and their product is

$(-180), i.e., 18 \times (-10).$

Clearly, $15 + (-12) = 3$ and $15 \times (-12) = -180.$

$$= 18x^2 + 15x - 12x - 10$$

$$= 3x(6x + 5) - 2(6x + 5)$$

$$= (6x + 5)(3x - 2)$$

Solution 30:

$$2x^2 + 11x - 21$$

We have to split 11 into two numbers such that their sum is 11 and their product is

$(-42), i.e., 2 \times (-21).$

Clearly, $14 + (-3) = 11$ and $14 \times (-3) = -42$

$$= 2x^2 + 11x - 21$$

$$= 2x^2 + 14x - 3x - 21$$

$$= 2x(x + 7) - 3(x + 7)$$

$$= (x + 7)(2x - 3)$$

Solution 31:

$$15x^2 + 2x - 8$$

We have to split 2 into two numbers such that their sum is 2 and their product

is $15 \times (-8) = -120$

Clearly, $12 + (-10) = 2$ and $12 \times (-10) = -120$

$$= 15x^2 + 12x - 10x - 8$$

$$= 3x(5x + 4) - 2(5x + 4)$$

$$= (5x + 4)(3x - 2)$$

Solution 32:

$$21x^2 + 5x - 6$$

$$= 21x^2 + 14x - 9x - 6$$

$$\begin{aligned} &= 7x(3x + 2) - 3(3x + 2) \\ &= (3x + 2)(7x - 3) \end{aligned}$$

Solution 33:

$$\begin{aligned} &24x^2 - 41x + 12 \\ &= 24x^2 - 32x - 9x + 12 \\ &= 8x(3x - 4) - 3(3x - 4) \\ &= (3x - 4)(8x - 3) \end{aligned}$$

Solution 34:

$$\begin{aligned} &3x^2 - 14x + 8 \\ &= 3x^2 - 12x - 2x + 8 \\ &= 3x(x - 4) - 2(x - 4) \\ &= (x - 4)(3x - 2) \end{aligned}$$

Solution 35:

$$\begin{aligned} &2x^2 + 3x - 90 \\ &= 2x^2 - 12x + 15x - 90 \\ &= 2x(x - 6) + 15(x - 6) \\ &= (x - 6)(2x + 15) \end{aligned}$$

Solution 36:

$$\begin{aligned} &\sqrt{5}x^2 + 2x - 3\sqrt{5} \\ &= \sqrt{5}x^2 + 5x - 3x - 3\sqrt{5} \\ &= \sqrt{5}x(x + \sqrt{5}) - 3(x + \sqrt{5}) \\ &= (x + \sqrt{5})(\sqrt{5}x - 3) \end{aligned}$$

Solution 37:

$$2\sqrt{3}x^2 + x - 5\sqrt{3}$$

$$\begin{aligned} &= 2\sqrt{3}x^2 + 6x - 5x - 5\sqrt{3} \\ &= 2\sqrt{3}x(x + \sqrt{3}) - 5(x + \sqrt{3}) \\ &= (x + \sqrt{3})(2\sqrt{3}x - 5) \end{aligned}$$

Solution 38:

$$\begin{aligned} &7x^2 + 2\sqrt{14}x + 2 \\ &= 7x^2 + \sqrt{14}x + \sqrt{14}x + 2 \\ &= \sqrt{7}x(\sqrt{7}x + \sqrt{2}) + \sqrt{2}(\sqrt{7}x + \sqrt{2}) \\ &= (\sqrt{7}x + \sqrt{2})(\sqrt{7}x + \sqrt{2}) \\ &= (\sqrt{7}x + \sqrt{2})^2 \end{aligned}$$

Solution 39:

$$\begin{aligned} &6\sqrt{3}x^2 - 47x + 5\sqrt{3} \\ &= 6\sqrt{3}x^2 - 2x - 45x + 5\sqrt{3} \\ &= 2x(3\sqrt{3}x - 1) - 5\sqrt{3}(3\sqrt{3}x - 1) \\ &= (3\sqrt{3}x - 1)(2x - 5\sqrt{3}) \end{aligned}$$

Solution 40:

$$\begin{aligned} &5\sqrt{5}x^2 + 20x + 3\sqrt{5} \\ &= 5\sqrt{5}x^2 + 15x + 5x + 3\sqrt{5} \\ &= 5x(\sqrt{5}x + 3) + \sqrt{5}(\sqrt{5}x + 3) \\ &= (\sqrt{5}x + 3)(5x + \sqrt{5}) \end{aligned}$$

Solution 41:

$$\begin{aligned} &\sqrt{3}x^2 + 10x + 8\sqrt{3} \\ &= \sqrt{3}x^2 + 6x + 4x + 8\sqrt{3} \\ &= \sqrt{3}x(x + 2\sqrt{3}) + 4(x + 2\sqrt{3}) \\ &= (x + 2\sqrt{3})(\sqrt{3}x + 4) \end{aligned}$$

Solution 42:

$$\begin{aligned}\sqrt{2}x^2 + 3x + \sqrt{2} \\&= \sqrt{2}x^2 + 2x + x + \sqrt{2} \\&= \sqrt{2}x(x + \sqrt{2}) + (x + \sqrt{2}) \\&= (x + \sqrt{2})(\sqrt{2}x + 1)\end{aligned}$$

Solution 43:

$$\begin{aligned}2x^2 + 3\sqrt{3}x + 3 \\&= 2x^2 + 2\sqrt{3}x + \sqrt{3}x + 3 \\&= 2x(x + \sqrt{3}) + \sqrt{3}(x + \sqrt{3}) \\&= (x + \sqrt{3})(2x + \sqrt{3})\end{aligned}$$

Solution 44:

$$\begin{aligned}15x^2 - x - 28 \\&= 15x^2 - 21x + 20x - 28 \\&= 3x(5x - 7) + 4(5x - 7) \\&= (5x - 7)(3x + 4)\end{aligned}$$

Solution 45:

$$\begin{aligned}6x^2 - 5x - 21 \\&= 6x^2 + 9x - 14x - 21 \\&= 3x(2x + 3) - 7(2x + 3) \\&= (2x + 3)(3x - 7)\end{aligned}$$

Solution 46:

$$\begin{aligned}2x^2 - 7x - 15 \\&= 2x^2 - 10x + 3x - 15 \\&= 2x(x - 5) + 3(x - 5) \\&= (x - 5)(2x + 3)\end{aligned}$$

Solution 47:

$$\begin{aligned} & 5x^2 - 16x - 21 \\ &= 5x^2 + 5x - 21x - 21 \\ &= 5x(x + 1) - 21(x + 1) \\ &= (x + 1)(5x - 21) \end{aligned}$$

Solution 48:

$$\begin{aligned} & 6x^2 - 11x - 35 \\ &= 6x^2 - 21x + 10x - 35 \\ &= 3x(2x - 7) + 5(2x - 7) \\ &= (2x - 7)(3x + 5) \end{aligned}$$

Solution 49:

$$\begin{aligned} & 9x^2 - 3x - 20 \\ &= 9x^2 - 15x + 12x - 20 \\ &= 3x(3x - 5) + 4(3x - 5) \\ &= (3x - 5)(3x + 4) \end{aligned}$$

Solution 50:

$$\begin{aligned} & 10x^2 - 9x - 7 \\ &= 10x^2 + 5x - 14x - 7 \\ &= 5x(2x + 1) - 7(2x + 1) \\ &= (2x + 1)(5x - 7) \end{aligned}$$

Solution 51:

$$\begin{aligned} & x^2 - 2x + \frac{7}{16} \\ &= \frac{1}{16}(16x^2 - 32x + 7) \\ &= \frac{1}{16}\{16x^2 - 4x - 28x + 7\} \end{aligned}$$

$$\begin{aligned} &= \frac{1}{16} \{4x(4x - 1) - 7(4x - 1)\} \\ &= \frac{1}{16} (4x - 1)(4x - 7) \end{aligned}$$

Solution 52:

$$\begin{aligned} &\frac{1}{3}x^2 - 2x - 9 \\ &= \frac{1}{3} \{x^2 - 6x - 27\} \\ &= \frac{1}{3} \{x^2 - 9x + 3x - 27\} \\ &= \frac{1}{3} \{x(x - 9) + 3(x - 9)\} \\ &= \frac{1}{3} (x - 9)(x + 3) \end{aligned}$$

Solution 53:

$$\begin{aligned} &x^2 + \frac{12}{35}x + \frac{1}{35} \\ &= \frac{1}{35} (35x^2 + 12x + 1) \\ &= \frac{1}{35} \{35x^2 + 7x + 5x + 1\} \\ &= \frac{1}{35} \{7x(5x + 1) + (5x + 1)\} \\ &= \frac{1}{35} (7x + 1)(5x + 1) \\ &= \left(\frac{7x}{35} + \frac{1}{35}\right) (5x + 1) \\ &= (5x + 1) \left(\frac{x}{5} + \frac{1}{35}\right) \end{aligned}$$

Solution 54:

$$21x^2 - 2x + \frac{1}{21}$$

$$\begin{aligned} &= 21x^2 - x - x + \frac{1}{21} \\ &= 21x\left(x - \frac{1}{21}\right) - \left(x - \frac{1}{21}\right) \\ &= \left(x - \frac{1}{21}\right)(21x - 1) \end{aligned}$$

Solution 55:

$$\begin{aligned} &\frac{3}{2}x^2 + 16x + 10 \\ &= \frac{3}{2}x^2 + 15x + x + 10 \\ &= \frac{3}{2}x(x + 10) + (x + 10) \\ &= (x + 10)\left(\frac{3}{2}x + 1\right) \end{aligned}$$

Solution 56:

$$\begin{aligned} &\frac{2}{3}x^2 - \frac{17}{3}x - 28 \\ &= \frac{2}{3}x^2 - 8x + \frac{7}{3}x - 28 \\ &= 2x\left(\frac{1}{3}x - 4\right) + 7\left(\frac{1}{3}x - 4\right) \\ &= \left(\frac{1}{3}x - 4\right)(2x + 7) \end{aligned}$$

Solution 57:

$$\begin{aligned} &\frac{3}{5}x^2 - \frac{19}{5}x + 4 \\ &= \frac{3}{5}x^2 - 3x - \frac{4}{5}x + 4 \\ &= 3x\left(\frac{1}{5}x - 1\right) - 4\left(\frac{1}{5}x - 1\right) \\ &= \left(\frac{1}{5}x - 1\right)(3x - 4) \end{aligned}$$

Solution 58:

$$\begin{aligned}2x^2 - x + \frac{1}{8} \\&= 2x^2 - \frac{1}{2}x - \frac{1}{2}x + \frac{1}{8} \\&= 2x\left(x - \frac{1}{4}\right) - \frac{1}{2}\left(x - \frac{1}{4}\right) \\&= \left(x - \frac{1}{4}\right)\left(2x - \frac{1}{2}\right)\end{aligned}$$

Solution 59:

$$\begin{aligned}2(x + y)^2 - 9(x + y) - 5 \\&= 2(x + y)^2 - 10(x + y) + (x + y) - 5 \\&= 2(x + y)[(x + y) - 5] + \{(x + y) - 5\} \\&= \{(x + y) - 5\}\{2(x + y) + 1\} \\&= (x + y - 5)(2x + 2y + 1)\end{aligned}$$

Note: you can also solve this by considering $x + y = z$, and then solving $2z^2 - 9z - 5$,

Solution 60:

$$\begin{aligned}9(2a - b)^2 - 4(2a - b) - 13 \\&= 9(2a - b)^2 + 9(2a - b) - 13(2a - b) - 13 \\&= 9(2a - b)\{(2a - b) + 1\} - 13\{(2a - b) + 1\} \\&= \{(2a - b) + 1\}\{9(2a - b) - 13\} \\&= (2a - b + 1)(18a - 9b - 13)\end{aligned}$$

Solution 61:

$$\begin{aligned}7(x - 2y)^2 - 25(x - 2y) + 12 \\&= 7(x - 2y)^2 - 21(x - 2y) - 4(x - 2y) + 12 \\&= 7(x - 2y)\{(x - 2y) - 3\} - 4\{(x - 2y) - 3\} \\&= \{(x - 2y) - 3\}\{7(x - 2y) - 4\} \\&= (x - 2y - 3)(7x - 14y - 4)\end{aligned}$$

Solution 62:

$$\begin{aligned} & 10\left(3x + \frac{1}{x}\right)^2 - \left(3x + \frac{1}{x}\right) - 3 \\ &= 10\left(3x + \frac{1}{x}\right)^2 - 6\left(3x + \frac{1}{x}\right) + 5\left(3x + \frac{1}{x}\right) - 3 \\ &= 2\left(3x + \frac{1}{x}\right)\left\{5\left(3x + \frac{1}{x}\right) - 3\right\} + \left\{5\left(3x + \frac{1}{x}\right) - 3\right\} \\ &= \left\{5\left(3x + \frac{1}{x}\right) - 3\right\}\left\{2\left(3x + \frac{1}{x}\right) + 1\right\} \\ &= \left(15x + \frac{5}{x} - 3\right)\left(6x + \frac{2}{x} + 1\right) \end{aligned}$$

Solution 63:

$$\begin{aligned} & 6\left(2x - \frac{3}{x}\right)^2 + 7\left(2x - \frac{3}{x}\right) - 20 \\ &= 6\left(2x - \frac{3}{x}\right)^2 + 15\left(2x - \frac{3}{x}\right) - 8\left(2x - \frac{3}{x}\right) - 20 \\ &= 3\left(2x - \frac{3}{x}\right)\left\{2\left(2x - \frac{3}{x}\right) + 5\right\} - 4\left\{2\left(2x - \frac{3}{x}\right) + 5\right\} \\ &= \left\{2\left(2x - \frac{3}{x}\right) + 5\right\}\left\{3\left(2x - \frac{3}{x}\right) - 4\right\} \\ &= \left(4x - \frac{6}{x} + 5\right)\left(6x - \frac{9}{x} - 4\right) \end{aligned}$$

Solution 64:

$$\begin{aligned} & (a + 2b)^2 + 101(a + 2b) + 100 \\ &= (a + 2b)^2 + 100(a + 2b) + (a + 2b) + 100 \\ &= (a + 2b)\{(a + 2b) + 100\} + \{(a + 2b) + 100\} \\ &= \{(a + 2b) + 100\}\{(a + 2b) + 1\} \\ &= (a + 2b + 100)(a + 2b + 1) \end{aligned}$$

Solution 65:

$$4x^4 + 7x^2 - 2$$

$$\begin{aligned} &= 4x^4 + 8x^2 - x^2 - 2 \\ &= 4x^2(x^2 + 2) - 1(x^2 + 2) \\ &= (4x^2 - 1)(x^2 + 2) \end{aligned}$$

Solution 66:

$$\begin{aligned} &(999)^2 - 1^2 \\ &= (999 + 1)(999 - 1) \\ &= 1000 \times 998 \\ &= 998000 \end{aligned}$$

EXERCISE 3D

Solution 1:

$$\begin{aligned} \text{(i)} \quad & (a + 2b + 5c)^2 \\ &= a^2 + (2b)^2 + (5c)^2 + 2 \times a \times 2b + 2 \times 2b \times 5c + 2 \times 5c \times a \\ &= a^2 + 4b^2 + 25c^2 + 4ab + 20bc + 10ac \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad & (2a - b + c)^2 \\ &= (2a)^2 + (-b)^2 + c^2 + 2 \times 2a \times (-b) + 2 \times (-b) \times c + 2ca \\ &= 4a^2 + b^2 + c^2 - 4ab - 2bc + 4ac \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad & (a - 2b - 3c)^2 \\ &= a^2 + (-2b)^2 + (-3c)^2 + 2 \times a \times (-2b) + 2 \times (-2b) \times 3c + 2 \times (-3c) \times a \\ &= a^2 + 4b^2 + 9c^2 - 4ab + 12bc - 6ac \end{aligned}$$

Solution 2:

$$\begin{aligned} \text{(i)} \quad & (2a - 5b - 7c)^2 \\ &= \{(2a) + (-5b) + (-7c)\}^2 \\ &= (2a)^2 + (-5b)^2 + (-7c)^2 + 2(2a) \times (-5b) + 2(-5b) \times (-7c) + 2(-7c) \times (2a) \\ &= 4a^2 + 25b^2 + 49c^2 - 20ab + 70bc - 28ac \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad & (-3a + 4b - 5c)^2 \\ &= (-3a)^2 + (4b)^2 + (-5c)^2 + 2 \times (-3a) \times 4b + 2 \times 4b \times (-5c) + 2 \times (-5c) \times (-3a) \\ &= 9a^2 + 16b^2 + 25c^2 - 24ab - 40bc + 30ca \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad & \left(\frac{1}{2}a - \frac{1}{4}b + 2\right)^2 \\ &= \left\{\left(\frac{a}{2}\right) + \left(-\frac{b}{4}\right) + 2\right\}^2 \\ &= \left(\frac{a}{2}\right)^2 + \left(-\frac{b}{4}\right)^2 + 2^2 + 2 \times \left(\frac{a}{2}\right) \times \left(-\frac{b}{4}\right) + 2 \times \left(-\frac{b}{4}\right) \times 2 + 2 \times 2 \times \frac{a}{2} \end{aligned}$$

$$= \frac{a^2}{4} + \frac{b^2}{16} + 4 - \frac{ab}{4} - b + 2a$$

Solution 3:

$$\begin{aligned} & 4x^2 + 9y^2 + 16z^2 + 12xy - 24yz - 16xz \\ &= (2x)^2 + (3y)^2 + (-4z)^2 + 2 \times 2x \times 3y + 2 \times 3y \times (-4z) + 2 \times (-4z) \times 2x \\ &= (2x + 3y - 4z)^2 \end{aligned}$$

Solution 4:

$$\begin{aligned} & 9x^2 + 16y^2 + 4z^2 - 24xy + 16yz - 12xz \\ &= (-3x)^2 + (4y)^2 + (2z)^2 + 2 \times (-3x)(4y) + 2(4y)(2z) + 2(2z)(-3x) \\ &= (-3x + 4y + 2z)^2 \end{aligned}$$

Solution 5:

$$\begin{aligned} & 25x^2 + 4y^2 + 9z^2 - 20xy - 12yz + 30xz \\ &= (5x)^2 + (-2y)^2 + (3z)^2 + 2 \times 5x \times (-2y) + 2 \times (-2y) \times (3z) + 2 \times 3z \times 5x \\ &= (5x - 2y + 3z)^2 \end{aligned}$$

Solution 6:

$$\begin{aligned} & 16x^2 + 4y^2 + 9z^2 - 16xy - 12yz + 24xz \\ &= (4x)^2 + (-2y)^2 + (3z)^2 + 2 \times 4x \times (-2y) + 2 \times (-2y) \times 3z + 2 \times 3z \times 4x \\ &= (4x - 2y + 3z)^2 \end{aligned}$$

Solution 7:

$$\begin{aligned} & \text{(i) } (99)^2 \\ &= (100 - 1)^2 \\ &= \{100 + (-1)\}^2 \\ &= 100^2 + 2 \times 100 \times (-1) + (-1)^2 \\ &= 10000 - 200 + 1 \\ &= 9801 \end{aligned}$$

$$(ii) (995)^2$$

$$= (1000 - 5)^2$$

$$= (1000)^2 - 2 \times 1000 \times 5 + (-5)^2$$

$$= 1000000 - 10000 + 25$$

$$= 990025$$

$$(iii) (107)^2$$

$$= (100 + 7)^2$$

$$= (100)^2 + 2 \times 100 \times 7 + 7^2$$

$$= 10000 + 1400 + 49$$

$$= 11449$$

Exercise 3E

Solution 1: We know $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$

$$\begin{aligned} \text{(i)} \quad & (3x + 2)^3 \\ &= (3x)^3 + (2)^3 + 3 \times 3x \times 2(3x + 2) \\ &= 27x^3 + 8 + 18x(3x + 2) \\ &= 27x^3 + 8 + 54x^2 + 36x \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad & \left(3a + \frac{1}{4b}\right)^3 \\ &= (3a)^3 + \left(\frac{1}{4b}\right)^3 + 3 \times 3a \times \frac{1}{4b} \left(3a + \frac{1}{4b}\right) \\ &= 27a^3 + \frac{1}{64b^3} + \frac{9a}{4b} \times 3a + \frac{9a}{4b} \times \frac{1}{4b} \\ &= 27a^3 + \frac{1}{64b^3} + \frac{27a^2}{4b} + \frac{9a}{16b^2} \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad & \left(1 + \frac{2}{3}a\right)^3 \\ &= (1)^3 + \left(\frac{2}{3}a\right)^3 + 3 \times 1 \times \frac{2}{3}a \left(1 + \frac{2}{3}a\right) \\ &= 1 + \frac{8}{27}a^3 + 2 \left(1 + \frac{2}{3}a\right) \\ &= 1 + \frac{8}{27}a^3 + 2a \times 1 + 2a \times \frac{2}{3}a \\ &= 1 + \frac{8}{27}a^3 + 2a + \frac{4}{3}a^2 \\ &= \frac{8}{27}a^3 + \frac{4}{3}a^2 + 2a + 1 \end{aligned}$$

Solution 2:

We have $(a - b)^3 = a^3 - b^3 - 3ab(a - b)$

$$\begin{aligned} & \text{(i) } (5a - 3b)^3 \\ &= (5a)^3 - (3b)^3 - 3 \times 5a \times 3b(5a - 3b) \\ &= 125a^3 - 27b^3 - 45ab(5a - 3b) \\ &= 125a^3 - 27b^3 - 45ab \times 5a + 45ab \times 3b \\ &= 125a^3 - 27b^3 - 225a^2b + 135ab^2 \end{aligned}$$

$$\begin{aligned} & \text{(ii) } \left(3x - \frac{5}{x}\right)^3 \\ &= (3x)^3 - \left(\frac{5}{x}\right)^3 - 3 \times 3x \times \frac{5}{x} \left(3x - \frac{5}{x}\right) \\ &= 27x^3 - \frac{125}{x^3} - 45 \left(3x - \frac{5}{x}\right) \\ &= 27x^3 - \frac{125}{x^3} - 135x + \frac{225}{x} \end{aligned}$$

$$\begin{aligned} & \text{(iii) } \left(\frac{4}{5}a - 2\right)^3 \\ &= \left(\frac{4}{5}a\right)^3 - (2)^3 - 3 \times \frac{4}{5}a \times 2 \left(\frac{4}{5}a - 2\right) \\ &= \frac{64}{125}a^3 - 8 - \frac{24}{5}a \left(\frac{4}{5}a - 2\right) \\ &= \frac{64}{125}a^3 - 8 - \frac{24}{5}a \times \frac{4}{5}a + \frac{24}{5}a \times 2 \\ &= \frac{64}{125}a^3 - 8 - \frac{96a^2}{25} + \frac{48}{5}a \end{aligned}$$

Solution 3:

$$\begin{aligned} & 8a^3 + 27b^3 + 36a^2b + 54ab^2 \\ &= (2a)^3 + (3b)^3 + 3 \times (2a)^2 \times 3b + 3 \times 2a \times (3b)^2 \\ &= (2a + 3b)^3 \quad [\because (a + b)^3 = a^3 + b^3 + 3a^2b + 3ab^2] \end{aligned}$$

Solution 4:

$$64a^3 - 27b^3 - 144a^2b + 108ab^2$$

$$\begin{aligned} &= (4a)^3 - (3b)^3 - 3 \times (4a)^2 \times 3b + 3 \times 4a \times (3b)^2 \\ &= (4a - 3b)^3 \quad [\because (a - b)^3 = a^3 - b^3 - 3a^2b + 3ab^2] \end{aligned}$$

Solution 5:

$$\begin{aligned} &1 + \frac{27}{125}a^3 + \frac{9a}{5} + \frac{27a^2}{25} \\ &= 1^3 + \left(\frac{3}{5}a\right)^3 + 3 \times 1^2 \times \frac{3}{5}a + 3 \times 1 \times \left(\frac{3}{5}a\right)^2 \\ &= \left(1 + \frac{3}{5}a\right)^3 \end{aligned}$$

Solution 6:

$$\begin{aligned} &125x^3 - 27y^3 - 225x^2y + 135xy^2 \\ &= (5x)^3 - (3y)^3 - 3 \times (5x)^2 \times 3y + 3 \times 5x \times (3y)^2 \\ &= (5x - 3y)^3 \end{aligned}$$

Solution 7:

$$\begin{aligned} &a^3x^3 - 3a^2bx^2 + 3ab^2x - b^3 \\ &= (ax)^3 - 3 \times (ax)^2 \times b + 3 \times ax \times b^2 - b^3 \\ &= (ax - b)^3 \end{aligned}$$

Solution 8:

$$\begin{aligned} &\frac{64}{125}a^3 - \frac{96}{25}a^2 + \frac{48}{5}a - 8 \\ &= \left(\frac{4}{5}a\right)^3 - 3 \times \left(\frac{4}{5}a\right)^2 \times 2 + 3 \times \frac{4}{5}a \times 2^2 - 2^3 \\ &= \left(\frac{4}{5}a - 2\right)^3 \end{aligned}$$

Solution 9:

$$\begin{aligned} &a^3 - 12a(a - 4) - 64 \\ &= a^3 - 12a^2 + 48a - 64 \end{aligned}$$

$$= a^3 - 4^3 - 3 \times a^2 \times 4 + 3 \times a \times 4^2$$

$$= (a - 4)^3$$

Solution 10:

(i) $(103)^3$

$$= (100 + 3)^3$$

$$= (100)^3 + 3^3 + 3 \times (100)^2 \times 3 + 3 \times 100 \times 3^2$$

$$= 1000000 + 27 + 90000 + 2700$$

$$= 1092727$$

(ii) $(99)^3$

$$= (100 - 1)^3$$

$$= (100)^3 - 1^3 - 3 \times (100)^2 \times 1 + 3 \times 100 \times 1^2$$

$$= 1000000 - 1 - 30000 + 300$$

$$= 970299$$

Exercise 3F

Solution 1:

We have $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$

$$\begin{aligned}x^3 + 27 &= (x)^3 + (3)^3 \\&= (x + 3)(x^2 - 3x + 3^2) \\&= (x + 3)(x^2 - 3x + 9)\end{aligned}$$

Solution 2:

We have $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$

$$\begin{aligned}27a^3 + 64b^3 &= (3a)^3 + (4b)^3 \\&= (3a + 4b)\{(3a)^2 - 3a \times 4b + (4b)^2\} \\&= (3a + 4b)\{9a^2 - 12ab + 16b^2\}\end{aligned}$$

Solution 3:

$$\begin{aligned}125a^3 + \frac{1}{8} &= (5a)^3 + \left(\frac{1}{2}\right)^3 \\&= \left(5a + \frac{1}{2}\right)\left\{(5a)^2 - 5a \times \frac{1}{2} + \left(\frac{1}{2}\right)^2\right\} \\&= \left(5a + \frac{1}{2}\right)\left\{25a^2 - \frac{5a}{2} + \frac{1}{4}\right\}\end{aligned}$$

Solution 4:

$$\begin{aligned}216x^3 + \frac{1}{125} &= (6x)^3 + \left(\frac{1}{5}\right)^3\end{aligned}$$

$$\begin{aligned} &= \left(6x + \frac{1}{5}\right) \left\{ (6x)^2 - 6x \times \frac{1}{5} + \left(\frac{1}{5}\right)^2 \right\} \\ &= \left(6x + \frac{1}{5}\right) \left(36x^2 - \frac{6x}{5} + \frac{1}{25}\right) \end{aligned}$$

Solution 5:

$$\begin{aligned} &16x^4 + 54x \\ &= 2x(8x^3 + 27) \\ &= 2x\{(2x)^3 + 3^3\} \\ &= 2x(2x + 3)\{(2x)^2 - 2x \times 3 + 3^2\} \\ &= 2x(2x + 3)(4x^2 - 6x + 9) \end{aligned}$$

Solution 6:

$$\begin{aligned} &7a^3 + 56b^3 \\ &= 7(a^3 + 8b^3) \\ &= 7\{a^3 + (2b)^3\} \\ &= 7(a + 2b)\{a^2 - a \times 2b + (2b)^2\} \\ &= 7(a + 2b)(a^2 - 2ab + 4b^2) \end{aligned}$$

Solution 7:

$$\begin{aligned} &x^5 + x^2 \\ &= x^2(x^3 + 1) \\ &= x^2(x + 1)(x^2 - x + 1) \end{aligned}$$

Solution 8:

$$\begin{aligned} &a^3 + 0.008 \\ &= a^3 + (0.2)^3 \\ &= (a + 0.2)\{a^2 - 0.2a + (0.2)^2\} \\ &= (a + 0.2)(a^2 - 0.2a + 0.04) \end{aligned}$$

Solution 9:

We have $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

$$\begin{aligned}1 - 27a^3 &= 1^3 - (3a)^3 \\&= (1 - 3a)\{1 + 3a + (3a)^2\} \\&= (1 - 3a)(1 + 3a + 9a^2)\end{aligned}$$

Solution 10:

$$\begin{aligned}64x^3 - 343 &= (4x)^3 - 7^3 \\&= (4x - 7)\{(4x)^2 + 4x \times 7 + 7^2\} \\&= (4x - 7)(16x^2 + 28x + 49)\end{aligned}$$

Solution 11:

$$\begin{aligned}x^3 - 512 &= x^3 - 8^3 \\&= (x - 8)\{x^2 + 8x + 8^2\} \\&= (x - 8)(x^2 + 8x + 64)\end{aligned}$$

Solution 12:

$$\begin{aligned}a^3 - 0.064 &= a^3 - (0.4)^3 \\&= (a - 0.4)\{a^2 + 0.4a + (0.4)^2\} \\&= (a - 0.4)(a^2 + 0.4a + 0.16)\end{aligned}$$

Solution 13:

$$\begin{aligned}8x^3 - \frac{1}{27y^3} &= (2x)^3 - \left(\frac{1}{3y}\right)^3\end{aligned}$$

$$\begin{aligned} &= \left(2x - \frac{1}{3y}\right)(2x)^2 - \left(\frac{1}{3y}\right)^2 \\ &= \left(2x - \frac{1}{3y}\right)\left\{(2x)^2 + 2x \times \frac{1}{3y} + \left(\frac{1}{3y}\right)^2\right\} \\ &= \left(2x - \frac{1}{3y}\right)\left(4x^2 + \frac{2x}{3y} + \frac{1}{9y^2}\right) \end{aligned}$$

Solution 14:

$$\begin{aligned} &\frac{x^3}{216} - 8y^3 \\ &= \left(\frac{x}{6}\right)^3 - (2y)^3 \\ &= \left(\frac{x}{6} - 2y\right)\left(\frac{x}{6}\right)^2 - (2y)^2 \\ &= \left(\frac{x}{6} - 2y\right)\left\{\left(\frac{x}{6}\right)^2 + \frac{x}{6} \times 2y + (2y)^2\right\} \\ &= \left(\frac{x}{6} - 2y\right)\left\{\frac{x^2}{36} + \frac{xy}{3} + 4y^2\right\} \end{aligned}$$

Solution 15:

$$\begin{aligned} &x - 8xy^3 \\ &= x(1 - 8y^3) \\ &= x\{1^3 - (2y)^3\} \\ &= x(1 - 2y)(1 + 2y + (2y)^2) \\ &= x(1 - 2y)(1 + 2y + 4y^2) \end{aligned}$$

Solution 16:

$$\begin{aligned} &= 32x^4 - 500x \\ &= 4x(8x^3 - 125) \\ &= 4x\{(2x)^3 - 5^3\} \\ &= 4x(2x - 5)\{(2x)^2 + 2x \times 5 + 5^2\} \\ &= 4x(2x - 5)(4x^2 + 10x + 25) \end{aligned}$$

Solution 17:

$$\begin{aligned} & 3a^7b - 81a^4b^4 \\ &= 3a^4b(a^3 - 27b^3) \\ &= 3a^4b\{a^3 - (3b)^3\} \\ &= 3a^4b(a - 3b)\{a^2 + 3ab + 9b^2\} \end{aligned}$$

Solution 18:

$$\begin{aligned} & x^4y^4 - xy \\ &= xy(x^3y^3 - 1) \\ &= xy\{(xy)^3 - 1\} \\ &= xy(xy - 1)(x^2y^2 + xy + 1) \end{aligned}$$

Solution 19:

$$\begin{aligned} &= 8x^2y^3 - x^5 \\ &= x^2(8y^3 - x^3) \\ &= x^2\{(2y)^3 - x^3\} \\ &= x^2(2y - x)(4y^2 + x^2 + 2xy) \end{aligned}$$

Solution 20:

$$\begin{aligned} & 1029 - 3x^3 \\ &= 1029 - 3x^3 \\ &= 3(343 - x^3) \\ &= 3(7^3 - x^3) \\ &= 3(7 - x)(7^2 + 7x + x^2) \\ &= 3(7 - x)(49 + 7x + x^2) \end{aligned}$$

Solution 21:

$$\begin{aligned} & x^6 - 729 \\ &= (x^2)^3 - (9)^3 \\ &= (x^2 - 9)\{(x^2)^2 + x^2 \times 9 + 9^2\} \end{aligned}$$

$$\begin{aligned} &= (x^2 - 3^2)\{x^4 + 9x^2 + 81\} \\ &= (x + 3)(x - 3)(x^4 + 9x^2 + 81) \end{aligned}$$

Solution 22:

$$\begin{aligned} &x^9 - y^9 \\ &= (x^3)^3 - (y^3)^3 \\ &= \{x^3 - y^3\}\{(x^3)^2 + x^3y^3 + (y^3)^2\} \\ &= (x - y)(x^2 + xy + y^2)(x^6 + x^3y^3 + y^6) \end{aligned}$$

Solution 23:

$$\begin{aligned} &(a + b)^3 - (a - b)^3 \\ &= \{(a + b) - (a - b)\}\{(a + b)^2 + (a + b)(a - b) + (a - b)^2\} \\ &= (a + b - a + b)\{a^2 + 2ab + b^2 + a^2 - b^2 + a^2 - 2ab + b^2\} \\ &= 2b(3a^2 + b^2) \end{aligned}$$

Solution 24:

$$\begin{aligned} &8a^3 - b^3 - 4ax + 2bx \\ &= (2a)^3 - b^3 - 2x(2a - b) \\ &= (2a - b)\{(2a)^2 + 2ab + b^2\} - 2x(2a - b) \\ &= (2a - b)\{4a^2 + 2ab + b^2 - 2x\} \end{aligned}$$

Solution 25:

$$\begin{aligned} &a^3 + 3a^2b + 3ab^2 + b^3 - 8 \\ &= (a + b)^3 - 2^3 \\ &= (a + b - 2)\{(a + b)^2 + 2(a + b) + 2^2\} \\ &= (a + b - 2)\{(a + b)^2 + 2(a + b) + 4\} \end{aligned}$$

Solution 26:

$$a^3 - \frac{1}{a^3} - 2a + \frac{2}{a}$$

$$\begin{aligned}
 &= \left(a^3 - \frac{1}{a^3}\right) - 2\left(a - \frac{1}{a}\right) \\
 &= \left(a - \frac{1}{a}\right)\left(a^2 + 1 + \frac{1}{a^2}\right) - 2\left(a - \frac{1}{a}\right) \\
 &= \left(a - \frac{1}{a}\right)\left(a^2 + 1 + \frac{1}{a^2} - 2\right) \\
 &= \left(a - \frac{1}{a}\right)\left(a^2 + \frac{1}{a^2} - 1\right)
 \end{aligned}$$

Solution 27:

$$\begin{aligned}
 &2a^3 + 16b^3 - 5a - 10b \\
 &= 2\{a^3 + 8b^3\} - 5(a + 2b) \\
 &= 2\{a^3 + (2b)^3\} - 5(a + 2b) \\
 &= 2\{(a + 2b)(a^2 - 2ab + 4b^2)\} - 5(a + 2b) \quad [\because x^3 + y^3 = (x + y)(x^2 - xy + y^2)] \\
 &= (a + 2b)(2a^2 - 4ab + 8b^2) - 5(a + 2b) \\
 &= (a + 2b)(2a^2 - 4ab + 8b^2 - 5)
 \end{aligned}$$

Solution 28:

$$\begin{aligned}
 &a^6 + b^6 \\
 &= (a^2)^3 + (b^2)^3 \\
 &= (a^2 + b^2)\{(a^2)^2 - a^2b^2 + (b^2)^2\} \\
 &= (a^2 + b^2)(a^4 - a^2b^2 + b^4)
 \end{aligned}$$

Solution 29:

$$\begin{aligned}
 &a^{12} - b^{12} \\
 &= (a^6 + b^6)(a^6 - b^6) \\
 &= \{(a^2)^3 + (b^2)^3\}\{(a^3)^2 - (b^3)^2\} \\
 &= (a^2 + b^2)\{(a^2)^2 - a^2b^2 + (b^2)^2\}(a^3 + b^3)(a^3 - b^3) \\
 &= (a^2 + b^2)(a^4 - a^2b^2 + b^4)(a + b)(a^2 - ab + b^2)(a - b)(a^2 + ab + b^2) \\
 &= (a + b)(a - b)(a^2 + b^2)(a^2 + ab + b^2)(a^2 - ab + b^2)(a^4 - a^2b^2 + b^4)
 \end{aligned}$$

Solution 30:

$$\begin{aligned} & x^6 - 7x^3 - 8 \\ &= x^6 - 8x^3 + x^3 - 8 \\ &= x^3(x^3 - 8) + 1(x^3 - 8) \\ &= (x^3 - 8)(x^3 + 1) \\ &= (x^3 - 2^3)(x^3 + 1^3) \\ &= (x - 2)(x^2 + 2x + 2^2)(x + 1)(x^2 - x + 1) \\ &= (x - 2)(x + 1)(x^2 + 2x + 4)(x^2 - x + 1) \end{aligned}$$

Solution 31:

$$\begin{aligned} & x^3 - 3x^2 + 3x + 7 \\ &= x^3 - 3x^2 + 3x - 1 + 8 \\ &= (x^3 - 3 \times x^2 \times 1 + 3 \times x \times 1^2 - 1^3) + 8 \\ &= (x - 1)^3 + 2^3 \\ &= (x - 1 + 2)\{(x - 1)^2 - 2(x - 1) + 2^2\} \\ &= (x + 1)(x^2 - 2x + 1 - 2x + 2 + 4) \\ &= (x + 1)(x^2 - 4x + 7) \end{aligned}$$

Solution 32:

$$\begin{aligned} & (x + 1)^3 + (x - 1)^3 \\ &= \{(x + 1) + (x - 1)\}\{(x + 1)^2 - (x + 1)(x - 1) + (x - 1)^2\} \\ &= 2x\{x^2 + 2x + 1 - (x^2 - 1) + x^2 - 2x + 1\} \\ &= 2x(x^2 + 2x + 1 - x^2 + 1 + x^2 - 2x + 1) \\ &= 2x(x^2 + 3) \end{aligned}$$

Solution 33:

$$\begin{aligned} & (2a + 1)^3 + (a - 1)^3 \\ &= \{(2a + 1) + (a - 1)\}\{(2a + 1)^2 - (2a + 1)(a - 1) + (a - 1)^2\} \\ &= 3a\{4a^2 + 4a + 1 - 2a^2 + 2a - a + 1 + a^2 - 2a + 1\} \\ &= 3a\{3a^2 + 3a + 3\} \end{aligned}$$

$$= 9a(a^2 + a + 1)$$

Solution 34:

$$\begin{aligned} & 8(x + y)^3 - 27(x - y)^3 \\ &= \{2(x + y)\}^3 - \{3(x - y)\}^3 \\ &= \{2(x + y) - 3(x - y)\}[\{2(x + y)\}^2 + 2(x + y) \times 3(x - y) + \{3(x - y)\}^2] \\ &= \{2x + 2y - 3x + 3y\}[4(x^2 + 2xy + y^2) + 6(x^2 - y^2) + 9(x^2 - 2xy + y^2)] \\ &= (5y - x)[4x^2 + 8xy + 4y^2 + 6x^2 - 6y^2 + 9x^2 - 18xy + 9y^2] \\ &= (5y - x)[19x^2 - 10xy + 7y^2] \end{aligned}$$

Solution 35:

$$\begin{aligned} & (x + 2)^3 + (x - 2)^3 \\ &= (x + 2 + x - 2)\{(x + 2)^2 - (x + 2) \times (x - 2) + (x - 2)^2\} \\ &= 2x(x^2 + 4x + 4 - x^2 + 4 + x^2 - 4x + 4) \\ &= 2x(x^2 + 12) \end{aligned}$$

Solution 36:

$$\begin{aligned} & (x + 2)^3 - (x - 2)^3 \\ &= (x + 2 - x + 2)\{(x + 2)^2 + (x + 2) \times (x - 2) + (x - 2)^2\} \\ &= 4(x^2 + 4x + 4 + x^2 - 4 + x^2 - 4x + 4) \\ &= 4(3x^2 + 4) \end{aligned}$$

Solution 37:

L. H. S :

$$\begin{aligned} & \frac{0.85 \times 0.85 \times 0.85 + 0.15 \times 0.15 \times 0.15}{0.85 \times 0.85 - 0.85 \times 0.15 + 0.15 \times 0.15} \\ &= \frac{(0.85)^3 + (0.15)^3}{(0.85)^2 - 0.85 \times 0.15 + (0.15)^2} \\ &= \frac{(0.85 + 0.15)\{(0.85)^2 - 0.85 \times 0.15 + (0.15)^2\}}{(0.85)^2 - 0.85 \times 0.15 + (0.15)^2} \\ &= (0.85 + 0.15) \end{aligned}$$

$$= 1$$

$$= \text{R. H. S}$$

Solution 38:

L.H.S:

$$\frac{59 \times 59 \times 59 - 9 \times 9 \times 9}{59 \times 59 + 59 \times 9 + 9 \times 9}$$

$$= \frac{(59)^3 - 9^3}{(59)^2 + 59 \times 9 + 9^2}$$

$$= \frac{(59 - 9)\{(59)^2 + 59 \times 9 + 9^2\}}{(59)^2 + 59 \times 9 + 9^2}$$

$$= (59 - 9)$$

$$= 50$$

$$= \text{R.H.S}$$

Exercise 3G

Solution 1:

$$\begin{aligned} & (x + y - z)(x^2 + y^2 + z^2 - xy + yz + zx) \\ &= (x + y + (-z))[x^2 + y^2 + (-z^2) - xy + y(-z) - (-z)x] \end{aligned}$$

We know:

$$(a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) = a^3 + b^3 + c^3 - 3abc$$

here $a = x, b = y, c = -z$

$$\begin{aligned} &= (x + y + (-z))[x^2 + y^2 + (-z^2) - xy + y(-z) - (-z)x] \\ &= x^3 + y^3 + (-z^3) - 3x \times y \times (-z) \\ &= x^3 + y^3 - z^3 + 3xyz \end{aligned}$$

Solution 2:

$$\begin{aligned} & (x - y - z)(x^2 + y^2 + z^2 + xy - yz + xz) \\ &= [x + (-y) + (-z)][x^2 + y^2 + z^2 + xy - yz + xz] \end{aligned}$$

We know:

$$(a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) = a^3 + b^3 + c^3 - 3abc$$

here $a = x, b = y, c = -z$

$$\begin{aligned} &= (x + (-y) + (-z))(x^2 + y^2 + z^2 + xy - yz + xz) \\ &= x^3 - y^3 - z^3 - 3xyz \end{aligned}$$

Solution 3:

$$\begin{aligned} & (x - 2y + 3)(x^2 + 4y^2 + 2xy - 3x + 6y + 9) \\ &= (x - 2y + 3)(x^2 + 4y^2 + 9 + 2xy + 6y - 3x) \end{aligned}$$

We know:

$$(a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) = a^3 + b^3 + c^3 - 3abc$$

here $a = x, b = -2y, c = 3$

$$\begin{aligned} &= [x + (-2y) + 3][x^2 + (-2y)^2 + (3)^2 - x \times (-2y) - (-2y) \times 3 - 3 \times x] \\ &= (x)^3 + (-2y)^3 + (3)^3 - 3 \times (-2y) \times 3x \\ &= x^3 - 8y^3 + 27 + 18xy \end{aligned}$$

Solution 4:

$$(3x - 5y + 4)(9x^2 + 25y^2 + 15xy - 20y + 12x + 16) \\ = (3x + (-5y) + 4)(9x^2 + 25y^2 + 16 + 15xy - 20y + 12x)$$

We know:

$$(a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) = a^3 + b^3 + c^3 - 3abc$$

$$\text{Here, } a = 3x, b = -5y, c = 4$$

$$= (3x + (-5y) + 4)[(3x)^2 + (-5y)^2 + (4)^2 - 3x \times (-5y) - (-5y) \times 4 + 3x \times 4] \\ = (3x)^3 + (-5y)^3 + 4^3 - 3 \times 3x \times (-5y) \times 4 \\ = 27x^3 - 125y^3 + 64 + 180xy$$

Solution 5:

$$125a^3 + b^3 + 64c^3 - 60abc$$

$$= (5a)^3 + (b)^3 + (4c)^3 - 3 \times 5a \times b \times 4c$$

$$\text{We know, } a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$\text{Here } a = 5a, b = b, c = 4c$$

$$= (5a + b + 4c)[(5a)^2 + (b)^2 + (4c)^2 - 5a \times b - b \times 4c - 5a \times 4c] \\ = (5a + b + 4c)(25a^2 + b^2 + 16c^2 - 5ab - 4bc - 20ac)$$

Solution 6:

$$a^3 + 8b^3 + 64c^3 - 24abc$$

$$= a^3 + (2b)^3 + (4c)^3 - 3 \times a \times 2b \times 4c$$

$$\text{We know, } a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$\text{Here } a = a, b = 2b, c = 4c$$

$$= (a + 2b + 4c)[a^2 + (2b)^2 + (4c)^2 - a \times 2b - 2b \times 4c - 4c \times a] \\ = (a + 2b + 4c)(a^2 + 4b^2 + 16c^2 - 2ab - 8bc - 4ca)$$

Solution 7:

$$1 + b^3 + 8c^3 - 6bc$$

$$= (1)^3 + (b)^3 + (2c)^3 - 3 \times 1 \times b \times 2c$$

$$\text{We know, } a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

Here $a = 1, b = b, c = 2c$

$$\begin{aligned} &= (1 + b + 2c)[1^2 + b^2 + (2c)^2 - 1 \times b - b \times 2c - 1 \times 2c] \\ &= (1 + b + 2c)(1 + b^2 + 4c^2 - b - 2bc - 2c) \end{aligned}$$

Solution 8:

$$\begin{aligned} &216 + 27b^3 + 8c^3 - 108bc \\ &= (6)^3 + (3b)^3 + (2c)^3 - 3 \times 6 \times 3b \times 2c \end{aligned}$$

We know, $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$

Here $a = 6, b = 3b, c = 2c$

$$\begin{aligned} &= (6 + 3b + 2c)[6^2 + (3b)^2 + (2c)^2 - 6 \times 3b - 3b \times 2c - 2c \times 6] \\ &= (6 + 3b + 2c)(36 + 9b^2 + 4c^2 - 18b - 6bc - 12c) \end{aligned}$$

Solution 9:

$$\begin{aligned} &27a^3 - b^3 + 8c^3 + 18abc \\ &= (3a)^3 + (-b)^3 + (2c)^3 - 3 \times (3a) \times (-b) \times (2c) \end{aligned}$$

We know, $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$

Here $a = 3a, b = -b, c = 2c$

$$\begin{aligned} &= [3a + (-b) + 2c][(3a)^2 + (-b)^2 + (2c)^2 - 3a(-b) - (-b)2c - 3a \times 2c] \\ &= (3a - b + 2c)(9a^2 + b^2 + 4c^2 + 3ab + 2bc - 6ac) \end{aligned}$$

Solution 10:

$$\begin{aligned} &8a^3 + 125b^3 - 64c^3 + 120abc \\ &= (2a)^3 + (5b)^3 + (-4c)^3 - 3 \times (2a) \times (5b) \times (-4c) \end{aligned}$$

We know, $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$

Here $a = 2a, b = 5b, c = -4c$

$$\begin{aligned} &= (2a + 5b - 4c)[(2a)^2 + (5b)^2 + (-4c)^2 - (2a) \times (5b) - (5b) \times (-4c) \\ &\quad - (2a) \times (-4c)] \\ &= (2a + 5b - 4c)(4a^2 + 25b^2 + 16c^2 - 10ab + 20bc + 8ac) \end{aligned}$$

Solution 11:

$$8 - 27b^3 - 343c^3 - 126bc$$

$$= (2)^3 + (-3b)^3 + (-7c)^3 - 3 \times (2) \times (-3b) \times (-7c)$$

$$\text{We know, } a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$\text{Here } a = 2, b = -3b, c = -7c$$

$$= [2 + (-3b) + (-7c)][(2)^2 + (-3b)^2 + (-7c)^2 - (2)(-3b) - (-3b)(-7c) - (2)(-7c)]$$

$$= (2 - 3b - 7c)(4 + 9b^2 + 49c^2 + 6b - 21bc + 14c)$$

Solution 12:

$$125 - 8x^3 - 27y^3 - 90xy$$

$$= 5^3 + (-2x)^3 + (-3y)^3 - 3 \times 5 \times (-2x) \times (-3y)$$

$$\text{We know, } a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$\text{Here } a = 5, b = -2x, c = -3y$$

$$= [5 + (-2x) + (-3y)][5^2 + (-2x)^2 + (-3y)^2 - 5 \times (-2x) - (-2x)(-3y) - 5 \times (-3y)]$$

$$= (5 - 2x - 3y)(25 + 4x^2 + 9y^2 + 10x - 6xy + 15y)$$

Solution 13:

$$2\sqrt{2}a^3 + 16\sqrt{2}b^3 + c^3 - 12abc$$

$$= (\sqrt{2}a)^3 + (2\sqrt{2}b)^3 + c^3 - 3 \times (\sqrt{2}a) \times (2\sqrt{2}b) \times (c)$$

$$\text{We know, } a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$\text{Here } a = \sqrt{2}a, b = 2\sqrt{2}b, c = c$$

$$= (\sqrt{2}a + 2\sqrt{2}b + c)[(\sqrt{2}a)^2 + (2\sqrt{2}b)^2 + c^2 - (\sqrt{2}a) \times (2\sqrt{2}b) - (2\sqrt{2}b) \times (c) - (\sqrt{2}a) \times (c)]$$

$$= (\sqrt{2}a + 2\sqrt{2}b + c)(2a^2 + 8b^2 + c^2 - 4ab - 2\sqrt{2}bc - \sqrt{2}ac)$$

Solution 14:

$$27x^3 - y^3 - z^3 - 9xyz$$

$$= (3x)^3 - y^3 - z^3 - 3 \times (3x) \times (-y) \times (-z)$$

We know, $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$

Here, $a = 3x, b = -y, c = -z$

$$= (3x + (-y) + (-z))[(3x)^2 + y^2 + z^2 - 3x \times (-y) - (-y) \times (-z) - 3x \times (-z)]$$

$$= (3x - y - z)(9x^2 + y^2 + z^2 + 3xy - yz + 3xz)$$

Solution 15:

$$2\sqrt{2}a^3 + 3\sqrt{3}b^3 + c^3 - 3\sqrt{6}abc$$

$$= (\sqrt{2}a)^3 + (\sqrt{3}b)^3 + c^3 - 3(\sqrt{2}a)(\sqrt{3}b)c$$

We know, $x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)$

$$x = \sqrt{2}a, y = \sqrt{3}b, z = c$$

$$= (\sqrt{2}a)^3 + (\sqrt{3}b)^3 + c^3 - 3(\sqrt{2}a)(\sqrt{3}b)c$$

$$= (\sqrt{2}a + \sqrt{3}b + c)(2a^2 + 3b^2 + c^2 - \sqrt{6}ab - \sqrt{3}bc - \sqrt{2}ac)$$

Solution 16:

$$3\sqrt{3}a^3 - b^3 - 5\sqrt{5}c^3 - 3\sqrt{15}abc$$

$$= (\sqrt{3}a)^3 + (-b)^3 + (-\sqrt{5}c)^3 - 3(\sqrt{3}a)(-b)(-\sqrt{5}c)$$

We know, $x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)$

$$x = \sqrt{3}a, y = -b, z = -\sqrt{5}c$$

$$= (\sqrt{3}a)^3 + (-b)^3 + (-\sqrt{5}c)^3 - 3(\sqrt{3}a)(-b)(-\sqrt{5}c)$$

$$= (\sqrt{3}a + (-b) + (-\sqrt{5}c))\{(\sqrt{3}a)^2 + (-b)^2 + (-\sqrt{5}c)^2 - \sqrt{3}a(-b) - (-b)(-\sqrt{5}c) - (-\sqrt{5}c) \times \sqrt{3}a\}$$

$$= (\sqrt{3}a - b - \sqrt{5}c)(3a^2 + b^2 + 5c^2 + \sqrt{3}ab - \sqrt{5}bc + \sqrt{15}ac)$$

Solution 17:

$$(a - b)^3 + (b - c)^3 + (c - a)^3$$

Putting $(a - b) = x, (b - c) = y$ and $(c - a) = z,$

we get: $(a - b)^3 + (b - c)^3 + (c - a)^3$

$$= x^3 + y^3 + z^3$$

$$[\text{where } (x + y + z) = (a - b) + (b - c) + (c - a) = 0]$$

$$= 3xyz$$

$$[(x + y + z) = 0 \Rightarrow x^3 + y^3 + z^3 = 3xyz]$$

$$= 3(a - b)(b - c)(c - a)$$

Solution 18:

$$(a - 3b)^3 + (3b - c)^3 + (c - a)^3$$

$$\text{Now, } (a - 3b) + (3b - c) + (c - a)$$

$$= a - 3b + 3b - c + c - a$$

$$= 0$$

We know that if $x + y + z = 0$,

$$\text{Then } x^3 + y^3 + z^3 = 3xyz$$

Hence,

$$(a - 3b)^3 + (3b - c)^3 + (c - a)^3$$

$$= 3(a - 3b)(3b - c)(c - a)$$

Solution 19:

We have:

$$(3a - 2b) + (2b - 5c) + (5c - 3a) = 0$$

$$\text{So, } (3a - 2b)^3 + (2b - 5c)^3 + (5c - 3a)^3$$

$$= 3(3a - 2b)(2b - 5c)(5c - 3a).$$

Solution 20:

$$\text{Given equation is } (5a - 7b)^3 + (7b - 9c)^3 + (9c - 5a)^3$$

$$\text{Now, } (5a - 7b) + (7b - 9c) + (9c - 5a)$$

$$= 5a - 7b + 7b - 9c + 9c - 5a$$

$$= 0$$

We know that if $x + y + z = 0$, then $x^3 + y^3 + z^3 = 3xyz$

$$\text{Hence, } (5a - 7b)^3 + (7b - 9c)^3 + (9c - 5a)^3$$

$$= 3(5a - 7b)(7b - 9c)(9c - 5a)$$

Solution 21:

$$a^3(b-c)^3 + b^3(c-a)^3 + c^3(a-b)^3$$

$$= [a(b-c)]^3 + [b(c-a)]^3 + [c(a-b)]^3$$

$$\text{Now, since, } a(b-c) + b(c-a) + c(a-b)$$

$$= ab - ac + bc - ba + ca - bc$$

$$= 0$$

$$\text{So, } a^3(b-c)^3 + b^3(c-a)^3 + c^3(a-b)^3$$

$$= 3a(b-c)b(c-a)c(a-b)$$

$$= 3abc(a-b)(b-c)(c-a)$$

Solution 22:

$$(i) (-12)^3 + 7^3 + 5^3$$

$$\text{Now, } -12 + 7 + 5 = -12 + 12 = 0$$

$$\text{We know that if } x + y + z = 0, \text{ then } x^3 + y^3 + z^3 = 3xyz$$

$$\text{Hence, } (-12)^3 + 7^3 + 5^3$$

$$= 3(-12)(7)(5)$$

$$= -1260$$

$$(ii) (28)^3 + (-15)^3 + (-13)^3$$

$$\text{Now, } 28 + (-15) + (-13) = 28 - 28 = 0$$

$$\text{We know that if } x + y + z = 0, \text{ then } x^3 + y^3 + z^3 = 3xyz$$

$$\text{Hence, } (28)^3 + (-15)^3 + (-13)^3$$

$$= 3(28)(-15)(-13)$$

$$= 16380$$

Solution 23:

$$\text{L.H.S.} = (a+b+c)^3 - a^3 - b^3 - c^3$$

$$= [(a+b) + c]^3 - a^3 - b^3 - c^3$$

$$= (a+b)^3 + c^3 + 3(a+b)c \times [(a+b) + c] - a^3 - b^3 - c^3$$

$$= a^3 + b^3 + 3ab(a+b) + c^3 + 3(a+b)c \times [(a+b) + c] - a^3 - b^3 - c^3$$

$$= 3ab(a+b) + 3(a+b)c \times [(a+b) + c]$$

$$\begin{aligned} &= 3(a+b)[ab + c(a+b) + c^2] \\ &= 3(a+b)[ab + ac + bc + c^2] \\ &= 3(a+b)[a(b+c) + c(b+c)] \\ &= 3(a+b)[(b+c)(a+c)] \\ &= 3(a+b)(b+c)(c+a) \\ &= \text{R.H.S.} \end{aligned}$$

Solution 24:

$$\begin{aligned} \text{L. H. S.} &= \frac{a^2}{bc} + \frac{b^2}{ca} + \frac{c^2}{ab} \\ &= \frac{a^3 + b^3 + c^3}{abc} \\ &= \frac{3abc}{abc} \quad [\because a + b + c = 0; \therefore a^3 + b^3 + c^3 = 3abc] \\ &= 3 \\ &= \text{R.H.S} \end{aligned}$$

Solution 25:

$$\begin{aligned} a^3 + b^3 + c^3 - 3abc &= (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca) \\ &= (a+b+c)[(a^2 + b^2 + c^2) - (ab + bc + ca)] \dots(i) \end{aligned}$$

$$\text{Now, } (a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$$

$$\Rightarrow (9)^2 = 35 + 2(ab + bc + ca)$$

$$\Rightarrow 81 = 35 + 2(ab + bc + ca)$$

$$\Rightarrow 2(ab + bc + ca) = 46$$

$$\Rightarrow ab + bc + ca = 23$$

Substituting in (i), we get

$$a^3 + b^3 + c^3 - 3abc = (9)[35 - 23] = 9 \times 12 = 108$$

MCQ

Answer 1: (c) 2

$$\text{Let } p(x) = 2x^2 + kx$$

Since $(x + 1)$ is a factor of $p(x)$,

$$P(-1) = 0$$

$$\Rightarrow 2(-1)^2 + k(-1) = 0$$

$$\Rightarrow 2 - k = 0$$

$$\Rightarrow k = 2$$

Answer 2: (d) 497

$$(249)^2 - (248)^2$$

$$= (249 + 248)(249 - 248)$$

$$= 497 \times 1$$

$$= 497$$

Answer 3: (c) 0

$$\frac{x}{y} + \frac{y}{x} = -1$$

$$\Rightarrow \frac{x^2 + y^2}{xy} = -1$$

$$\Rightarrow x^2 + y^2 = -xy$$

$$\Rightarrow x^2 + y^2 + xy = 0$$

Now,

$$(x^3 - y^3) = (x - y)(x^2 + y^2 + xy)$$

$$= (x - y)(0)$$

$$= 0$$

Answer 4: (d) 3abc

We know that, if $x + y + z = 0$ then, $x^3 + y^3 + z^3 = 3xyz$

$$\therefore a^3 + b^3 + c^3 = 3abc$$

Answer 5: (c) $\frac{1}{4}$

$$\left(3x + \frac{1}{2}\right)\left(3x - \frac{1}{2}\right) = 9x^2 - p$$

$$\Rightarrow (3x)^2 - \left(\frac{1}{2}\right)^2 = 9x^2 - p$$

$$\Rightarrow 9x^2 - \frac{1}{4} = 9x^2 - p$$

$$\Rightarrow -\frac{1}{4} = -p$$

$$\Rightarrow p = \frac{1}{4}$$

Answer 6: (d) 27

$$\text{we know that, } (a + b)^3 = a^3 + b^3 + 3a^2b + 3ab^2$$

$$\begin{aligned}\text{so, } (x + 3)^3 &= x^3 + 3^3 + 3(x^2)(3) + 3x(3^2) \\ &= x^3 + 27 + 9x^2 + 27x\end{aligned}$$

Thus, the coefficient of x is 27

Answer 7: (d) $3xy$

$$\begin{aligned}(x + y)^3 - (x^3 + y^3) &= (x + y)^3 - (x + y)(x^2 - xy + y^2) \\ &= (x + y)[(x + y)^2 - (x^2 - xy + y^2)] \\ &= (x + y)[x^2 + 2xy + y^2 - x^2 + xy - y^2] \\ &= (x + y)(3xy)\end{aligned}$$

Answer 8: (d) $10x$

$$\begin{aligned}(25x^2 - 1) + (1 + 5x)^2 &= [(5x)^2 - (1)^2] + (1 + 5x)^2\end{aligned}$$

$$\begin{aligned} &= (5x - 1)(5x + 1) + (1 + 5x)^2 \\ &= (1 + 5x)[(5x - 1) + (1 + 5x)] \\ &= (1 + 5x)(5x - 1 + 1 + 5x) \\ &= (1 + 5x)(10x) \end{aligned}$$

Answer 9: (b) 5

Since $(x + 5)$ is a factor of $p(x) = x^3 - 20x + 5k$,

$$\begin{aligned} p(-5) &= 0 \\ \Rightarrow (-5)^3 - 20(-5) + 5k &= 0 \\ \Rightarrow -125 + 100 + 5k &= 0 \\ \Rightarrow -25 + 5k &= 0 \\ \Rightarrow 5k &= 25 \\ \Rightarrow k &= 5 \end{aligned}$$

Answer 10: (b) $m = 7, n = -18$

Let: $p(x) = x^3 + 10x^2 + mx + n$

Now,

$$x + 2 = 0$$

$$\Rightarrow x = -2$$

$\therefore (x + 2)$ is a factor of $p(x)$.

So, $p(-2) = 0$

$$\Rightarrow (-2)^3 + 10(-2)^2 + m(-2) + n = 0$$

$$\Rightarrow -8 + 40 - 2m + n = 0$$

$$\Rightarrow 2m - n = 32 \quad \dots (i)$$

Now,

$$x - 1 = 0$$

$$\Rightarrow x = 1$$

Also, $(x - 1)$ is a factor of $p(x)$.

$$\text{So, } p(1) = 0$$

$$\Rightarrow (1)^3 + 10(1)^2 + m(1) + n = 0$$

$$\Rightarrow 1 + 10 + m + n = 0$$

$$\Rightarrow m + n = -11 \quad \dots (ii)$$

Adding (i) and (ii), we get

$$3m = 21$$

$$\Rightarrow m = 7$$

Substituting in (ii), we get

$$n = -18$$

Answer 11: (b) 9984

$$104 \times 96$$

$$= (100 + 4)(100 - 4)$$

$$= 100^2 - 4^2$$

$$= 10000 - 16$$

$$= 9984$$

Answer 12: (c) 93940

$$305 \times 308$$

$$= (300 + 5) \times (300 + 8)$$

$$= 300 \times 300 + 300 \times 8 + 5 \times 300 + 5 \times 8$$

$$= 90000 + 2400 + 1500 + 40$$

$$= 93940$$

Answer 13: (b) 39951

$$207 \times 193$$

$$= (200 + 7)(200 - 7)$$

$$\begin{aligned} &= (200)^2 - (7)^2 \\ &= 40000 - 49 \\ &= 39951 \end{aligned}$$

Answer 14: (a) $(2a + b + 2)^2$

$$\begin{aligned} &4a^2 + b^2 + 4ab + 8a + 4b + 4 \\ &= 4a^2 + b^2 + 4 + 4ab + 4b + 8a \\ &= (2a)^2 + b^2 + 2^2 + 2(2a)b + 2b(2) + 2(2)(2a) \\ &= (2a + b + 2)^2 \quad [\because (x + y + z)^2 = x^2 + y^2 + z^2 + 2xy + 2yz + 2zx] \end{aligned}$$

Answer 15: (c) $(x - 7)(x + 3)$

$$\begin{aligned} &x^2 - 4x - 21 \\ &= x^2 - 7x + 3x - 21 \\ &= x(x - 7) + 3(x - 7) \\ &= (x - 7)(x + 3) \end{aligned}$$

Answer 16: (c) $(2x + 3)(2x - 1)$

$$\begin{aligned} &4x^2 + 4x - 3 \\ &= 4x^2 + 6x - 2x - 3 \\ &= 2x(2x + 3) - 1(2x + 3) \\ &= (2x + 3)(2x - 1) \end{aligned}$$

Answer 17: (b) $(2x + 5)(3x + 1)$

$$\begin{aligned} &6x^2 + 17x + 5 \\ &= 6x^2 + 15x + 2x + 5 \\ &= 3x(2x + 5) + 1(2x + 5) \\ &= (2x + 5)(3x + 1) \end{aligned}$$

Answer 18: (c) $x^3 + 2x^2 - x - 2$

Let: $f(x) = x^3 - 2x^2 + x + 2$

$x + 1 = 0 \Rightarrow x = -1$

By factor theorem, if $(x - a)$ is a factor of $f(x)$

then, $f(a) = 0$

replace x by -1 in each of the options, we get

option (a): $(-1)^3 - 2(-1)^2 + (-1) + 2 = -1 - 2 - 1 + 2 = -2$

option (b): $(-1)^3 + 2(-1)^2 + (-1) - 2 = -1 + 2 - 1 - 2 = -2$

option (c): $(-1)^3 + 2(-1)^2 - (-1) - 2 = -1 + 2 + 1 - 2 = 0$

option (d): $(-1)^3 + 2(-1)^2 - (-1) + 2 = -1 + 2 + 1 + 2 = 4$

Clearly, $(x+1)$ is a factor of the polynomial, $x^3 + 2x^2 - x - 2$

Answer 19: (d) $(3x + 2)(x^2 + 1)$

$3x^3 + 2x^2 + 3x + 2$

$= x^2(3x + 2) + 1(3x + 2)$

$= (3x + 2)(x^2 + 1)$

Answer 20: (d) 3

$a + b + c = 0 \Rightarrow a^3 + b^3 + c^3 = 3abc$

Thus, we have:

$$\left(\frac{a^2}{bc} + \frac{b^2}{ca} + \frac{c^2}{ab} \right) = \frac{a^3 + b^3 + c^3}{abc} = \frac{3abc}{abc} = 3$$

Answer 21: (a) 108

$x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)$

$= (x + y + z)[(x + y + z)^2 - 3(xy + yz + zx)]$

$= 9 \times (81 - 3 \times 23)$

$= 9 \times 12 = 108$

Answer 22: (d) 0

$$\frac{a}{b} + \frac{b}{a} = -1$$

$$\Rightarrow \frac{a^2 + b^2}{ab} = -1$$

$$\Rightarrow a^2 + b^2 = -ab$$

Thus, we have:

$$(a^3 - b^3) = (a - b)(a^2 + b^2 + ab)$$

$$= (a - b) \times (-ab + ab)$$

$$= (a - b) \times 0$$

$$= 0$$